

Closure-Aware Cut Certificates and First-Hit Factorization

Sharp homology budgets and terminal dimension bounds
for exact unknot recognition

A research contribution prepared for the ProveIt project
Mathematical analysis and reference implementation prepared with ChatGPT

8 October 2026

Abstract

We develop an exact, decision-oriented refinement of the graded and corridor transfer methods in ProveIt’s unknot-recognition project. A transferred cobordism differential is a finite typed path sum. Factoring that sum at the *first* visit to an arbitrary vertex separator gives an exact product $D = BA$, without an antichain hypothesis or commuting coefficients. After a fixed verified closure, the sum of the separator’s vector-space dimensions bounds the differential rank. We combine these bounds with the adjacent-degree constraints imposed by $d^2 = 0$. The resulting capacitated path-matching problem has a simple optimal greedy solution and yields the strongest homology lower bound available from those dimensions and rank caps alone.

For a certified reduced knot complex, a lower bound greater than one rejects the unknot without evaluating the remaining transferred coefficients. Otherwise the closed critical dimension is at most twice the optimized rank budget plus one. Exact rank computation through the cut factors avoids materializing a dense terminal differential. We prove the factorization, the sharp budget theorem, the decision dichotomy, and an explicit separation from both endpoint-propagation directions on a valid algebraic family.

The accompanying standard-library Python package passes 36 unit tests and an audit of 80 small braid diagrams in 160 matching configurations, covering 1,070 transferred maps. In seven-round paired experiments, min-cut discovery is overhead on the small knot examples; the largest constructed bottleneck example gives a $4.80\times$ exact-rank-stage improvement over forward propagation. This is not an end-to-end production speedup. The results provide an optional research kernel, independently replayable small-input certificates, and precise conditional quasi-polynomial bounds. They do not establish a quasi-polynomial bound for arbitrary knot diagrams or implement the missing geometric construction and production closure adapter.

Status. Complete proofs for the stated algebraic results; executed reference-code validation; proposed, not performed, integration into the maintained recognizer. No formal proof-assistant verification is claimed.

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1 Research target, provenance, and contribution

1.1 What is being improved

The practical question is not whether a full homology computation can be expressed as a finite calculation. It is whether the information needed for a *rank-one decision* can be extracted without creating every intermediate matrix entry or expanding every terminal generator. An exact recognition algorithm may reject as soon as certified homology dimension exceeds one; it need not compute that dimension in full.

The maintained project overview distinguishes the existing exact recognizers from a proved general quasi-polynomial implementation [7]. Two especially relevant maintained articles supply graded perturbation transfer and survivor-corridor pruning with forward or reverse propagation [8, 9]. We build on their path-sum viewpoint rather than reintroducing local cancellation, visible connected-sum decomposition, or an invariant filter as a new recognition algorithm.

The specific additional question addressed here is:

Can a small *internal* interface control an entire transferred map even when both terminal sets are large, and can that interface certify enough homology to make the remaining exact computation unnecessary?

The answer is yes for explicit acyclic transfer graphs, with closure-dependent capacities and a carefully stated knot-complex contract. The hard unresolved question is how cheaply such graphs and useful interfaces can be produced for arbitrary classical knot diagrams.

1.2 Repository snapshot and review scope

The branch reference observed during this work was

bc5b913850f78155ccfddd808d4b11517dca252d.

Its recorded commit time was 8 October 2026, 16:46:30 UTC. The reviewed full texts were the project overview and the maintained graded-transfer and corridor-transfer articles. Their Git blob identifiers are retained in `PROVENANCE.json`. The corridor article’s blob was additionally checked at the pinned commit. Directory listings and targeted repository searches were used to locate related work.

This is a source review, not a full checkout or a full upstream regression run. The supplied code is an independent research package: it neither vendors nor modifies the maintained scanner. In particular, the 36 tests reported here are this package’s tests, not the maintained project’s test count. Repository search was not an exhaustive novelty search. The elementary network and linear-algebra ingredients are not claimed as newly discovered mathematics; the contribution is their precise combination, specialization, proof obligations, and tested decision-oriented use at this transfer boundary.

1.3 External status of the asymptotic target

Oxford publicly announced Lackenby’s quasi-polynomial unknot-recognition result in 2021 [6]. His July 2026 hierarchy preprint gives a new algorithmic approach; Section 9 explicitly distinguishes a

bound on step counts from a bound on the cost of all individual steps [5]. An announcement of a general theorem, the specifications in that preprint, and the guarantees of this project’s executable recognizer are different claims. Our homological construction does not fill in those hierarchy routines, and no conclusion about their eventual complexity is inferred from our experiments.

1.4 Main statements and their logical order

The core results form a short dependency chain. First-hit factorization (Theorem 3.2) is purely categorical. Closure converts it into rank caps (Theorem 4.1). The optimal rank-budget theorem (Theorem 5.2) turns those caps into a sharp homology lower bound. The rank-one decision dichotomy (Theorem 6.1) then either rejects or bounds the size of the remaining closed terminal problem. Finally, Theorem 7.1 states the costs that must be bounded to obtain a quasi-polynomial recognition result on a specified input class.

The finite examples test implementations of these statements. They do not prove the geometric hypotheses of the complexity theorem. Conversely, negative timings on small diagrams do not invalidate the algebraic factorization or its asymptotic separation from endpoint propagation.

2 Exact input model and the transfer boundary

2.1 The knot-complex contract

Let K be a one-component classical knot and let $C(K)$ be its finite reduced Khovanov cochain complex over $\mathbb{F}_2$. We normalize the marked crossingless unknot to a single generator in bidegree $(0, 0)$. Thus a verified chain homotopy equivalent complex R has

$$\dim_{\mathbb{F}_2} H(R) = \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2).$$

The normalized degree matters for graded certificates, but total rank does not depend on an overall grading shift.

Kronheimer–Mrowka’s detection theorem gives the rank-one criterion [4]. For the binary version used here, universal coefficients imply that rational reduced rank is at most binary reduced dimension. The rational reduced theory is nonzero for a knot: its Euler polynomial is the normalized Jones polynomial, which takes value one at one. Hence binary dimension one forces rational rank one and therefore the unknot; the unknot itself has binary dimension one. A verified binary dimension exceeding one excludes the unknot simply by invariance. No assertion about detection by the Jones polynomial alone is used.

A graph with user-supplied dimensions is *not* a knot certificate. Any topological conclusion below requires that the graph describe an exact transferred differential for the specified reduced knot complex, or that a separate verified chain equivalence establish this fact. This input-origin condition is as important as the subsequent algebra.

2.2 Typed finite path sums

Let $\mathcal{A}$ be an additive category whose morphism groups are $\mathbb{F}_2$ -vector spaces and whose composition is bilinear. A finite directed acyclic graph $G = (V, E)$ has an object X_v at each vertex and a morphism $f_e : X_u \rightarrow X_v$ on each edge $e = (u, v)$. Let $S, T \subseteq V$ be disjoint source and target

sets. Sources have no incoming edges, and targets have no outgoing edges. Vertices unrelated to a source–target path may be retained; pruning them is optional.

For a path $\pi = (v_0, \dots, v_m)$ write

$$f_\pi = f_{v_{m-1}v_m} \circ \dots \circ f_{v_0v_1}.$$

A length-zero path contributes the identity. The path-sum transfer is

$$D_{ts} = \sum_{\pi: s \rightsquigarrow t} f_\pi, \quad D : \bigoplus_{s \in S} X_s \longrightarrow \bigoplus_{t \in T} X_t. \quad (1)$$

Acyclicity makes the sum finite. Distinct paths can cancel in characteristic two; nonzero individual morphisms can have zero composite. Reachability is therefore an upper approximation to algebraic support, not an exact test of nonzero entries.

The abstract theorem also works over another coefficient ring with an additive category and the appropriate signed transfer graph. The shipped integer encodings use XOR addition and so implement characteristic two only.

2.3 Why this graph appears in cancellation

Local cancellation in Khovanov computations and algebraic Morse theory are established constructions [1, 3, 10]. The particular production setting reviewed here splits $d = d_0 + \delta$, contracts the scalar part d_0 , and obtains

$$D = P\delta(1 + H\delta)^{-1}I = \sum_{r \geq 0} P\delta(H\delta)^r I. \quad (2)$$

Here I, P, H are the actual inclusion, projection, and contracting homotopy, not merely the ranks of the scalar matrices. Nilpotence follows from a verified acyclic order. In the maintained graded implementation that order uses the strictly increasing quantum weight of radical arrows [9]. In our closed reference implementation it is supplied by an explicitly checked acyclic Morse matching.

The degree- h transfer graph has copies of degree- h and degree- $(h + 1)$ objects, the terminal survivors, and edges for I, δ, H, P . The H arrows run back by one cohomological degree. Consecutive factors in (2) give exactly the directed source–target paths. The resulting path sum is a differential on the survivors because it comes from a chain contraction. An arbitrary path-sum matrix, in isolation, need not form a complex with its neighboring matrices.

Our closed cube tests use a simpler special case. Chosen scalar differential entries form disjoint identity pairs. Their arrows are reversed, all other differential arrows remain forward, and the whole directed graph is checked for cycles. The elementary two-term contractions and their finite perturbation produce the Morse differential. In this closed Morse model, unlike the production scalar/radical split, δ may itself contain scalar entries; acyclicity is the termination condition. All signs disappear over $\mathbb{F}_2$. This validation does not assume that a closed binary differential strictly raises the normalized quantum degree; in fact it preserves that degree.

3 First-hit factorization through an arbitrary separator

Definition 3.1. A vertex separator $Q \subseteq V$ meets every directed path from S to T . It may contain terminals, and it need not be an antichain. For $q \in Q$ define

$$A_{qs} = \sum_{\substack{\pi: s \rightsquigarrow q \\ q \text{ is the first vertex of } Q \text{ on } \pi}} f_\pi, \quad B_{tq} = \sum_{\rho: q \rightsquigarrow t} f_\rho.$$

The suffix sum is unrestricted: it may pass through other vertices of Q . If $s \in Q$, its only first-hit prefix is the length-zero path at s .

Theorem 3.2 (First-hit separator factorization). *In the typed acyclic setting of (1), every vertex separator Q gives*

$$D = BA : \bigoplus_{s \in S} X_s \xrightarrow{A} \bigoplus_{q \in Q} X_q \xrightarrow{B} \bigoplus_{t \in T} X_t. \quad (3)$$

No invertibility, commutativity, generic-rank, or antichain hypothesis is required. The factorization is valid even when the final sum vanishes.

Proof. Every source–target path meets Q and has a unique first such vertex q . Split it there. Its prefix is one of the paths contributing to A_{qs} , and its suffix contributes to B_{tq} . Conversely, concatenate any allowed prefix to q with any suffix from q to t . A repeated vertex other than q would form a directed cycle, so the concatenation is a path. Its first visit to Q is q . We have a bijection between original paths and the pairs of paths indexing the matrix product BA . Ordered composition of their morphisms gives the original path product. Bilinearity now proves each matrix entry of (3). Cancellations take place identically on both sides of the equality. $\square$

3.1 A tempting but incorrect variant

It is not valid to use unrestricted prefixes and unrestricted suffixes for an arbitrary separator. In the chain

$$s \longrightarrow q_1 \longrightarrow q_2 \longrightarrow t$$

put every edge coefficient equal to one and take $Q = \{q_1, q_2\}$. The original transfer is one. An unrestricted split at either cut vertex counts the same path twice, giving zero over $\mathbb{F}_2$. Thus an implementation that quietly treats every minimum vertex cut as an antichain can compute a wrong differential. The issue can occur for a minimum cut, not just a redundant separator. Add edges $s \rightarrow q_2$ and $q_1 \rightarrow t$ to the displayed chain, give s, t capacity three and q_1, q_2 capacity one. Then $\{q_1, q_2\}$ is the unique minimum vertex separator, of capacity two. There are three source–target paths, so the actual binary transfer is one; unrestricted splitting counts the two-cut path twice and again gives zero. First-hit prefixes remove the ambiguity without placing extra restrictions on the optimizer.

The same example explains why the suffixes must remain unrestricted. If both prefixes and suffixes were required to avoid other cut vertices, the unique original path would disappear. The two asymmetric rules are a pair: first-hit on the left, unrestricted on the right.

3.2 An exact implementation and its costs

Let $a = |S|$, $b = |T|$, and $r = |Q|$. For each q , compute its prefix row by reverse propagation from q , stopping propagation at every other vertex of Q . Contributions that reach another cut source are

explicitly excluded from the row. Compute its suffix column by ordinary forward propagation from q , without stopping at later cut vertices.

For an edge $u \xrightarrow{f} v$, reverse propagation of a current suffix $g : v \rightarrow q$ forms $g \circ f$. Forward propagation of $g : q \rightarrow u$ forms $f \circ g$. Exchanging those products is wrong in a typed category, even if a particular dot-ring test happens to commute. The matrix-category tests therefore use rectangular noncommuting binary matrices.

Proposition 3.3 (Factor construction bound). *After topological ordering, the factors can be computed in $O(r(|V| + |E|))$ coefficient/control operations, storing $O(|V| + |E| + r(a + b))$ coefficients and graph entries. This bound includes visits to vertices whose current coefficient is zero. The reference implementation's heap-based topological sort adds $O(|E| + |V| \log(|V| + 2))$ index operations.*

Proof. There are two propagations per cut vertex. Each visits every vertex once and attempts at most one composition and addition per edge. Working values are reused between roots. The permanent prefix and suffix arrays have ra and br entries. Standard adjacency construction and the stated sort supply the remaining terms. $\square$

This is a representation bound, not necessarily an improvement. If a minimum weighted cut contains many zero-dimensional objects, r can be large while its capacity is small. Zero-dimensional objects may be removed after verified closure, but an unclosed typed computation must not replace r by the capacity without a separate argument. Expanding BA naively costs up to abr additional compositions. The terminal rank query below avoids that expansion; a later tangle extension generally cannot.

4 Closure-weighted cuts and verifiable rank caps

4.1 The closure functor is part of the input

Fix an additive functor

$$F : \mathcal{A} \longrightarrow \text{Vect}_{\mathbb{F}_2}^{\text{fd}}.$$

For the intended tangle application, F is a specified marked, crossingless closure followed by the reduced Khovanov evaluation. It is not an unspecified future continuation containing crossings. We require that F preserve the relevant object shifts, the reduced marked structure, composition, and addition. A certificate for its action must be obtained from the established cobordism representation or a verified adapter.

Write $d_v = \dim F(X_v)$. Application of F to Theorem 3.2 gives ordinary linear maps through $W_Q = \bigoplus_{q \in Q} F(X_q)$.

Theorem 4.1 (Closure-capacity bound). *For every vertex separator Q ,*

$$\text{rank } F(D) \leq \kappa(Q) := \sum_{q \in Q} d_q. \quad (4)$$

The minimum of $\kappa(Q)$ is an exactly computable nonnegative-integer vertex-cut problem. A supplied separator alone certifies an upper bound; a feasible flow of the same value additionally certifies that the separator has minimum capacity.

Proof. The image of $F(D) = F(B)F(A)$ has dimension at most $\dim W_Q$. For optimization, split every vertex v into $v_{\text{in}}, v_{\text{out}}$ with an arc of capacity d_v . Replace each original graph edge by an arc from u_{out} to v_{in} of capacity $1 + \sum_v d_v$. Connect a super-source to original sources and original targets to a super-sink using that same large capacity. Cutting every vertex has cost at most $\sum_v d_v$, so no minimum cut uses a large-capacity arc. The remaining minimum cuts correspond exactly to vertex separators. Integral maximum flow and minimum cut agree. For verification, check each flow's capacity bounds and conservation, verify graph separation after deleting Q , and check equality of the flow value and $\kappa(Q)$. Weak duality then proves optimality, independently of the search routine. $\square$

Zero capacities are allowed. This is useful for a quantum-degree sector in which an object has no basis vectors. A single vertex of dimension d has capacity d , not one. In particular, a one-vertex separator is not a rank-one certificate unless the closure dimension of that vertex is at most one.

4.2 Dimensions of marked matching closures

Consider noncrossing matchings a, c on $2k > 0$ boundary points, with a fixed marked point. Their overlay is a disjoint union of circles; let $\ell = c(a, c)$. The marked circle contributes a one-dimensional reduced factor, and each other circle contributes the Frobenius algebra $\mathbb{F}_2[x]/(x^2)$ as a two-dimensional vector space. Therefore

$$d_a = 2^{\ell-1}. \quad (5)$$

Use degree zero for the normalized marked factor and degrees $+1, -1$ for $1, x$ on an unmarked circle. For an object with shift q , the dimension in quantum degree j is

$$d_{a,q}^j = \begin{cases} \binom{\ell-1}{(q+\ell-1-j)/2}, & (q+\ell-1-j)/2 \in \{0, \dots, \ell-1\}, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

The integer condition includes parity. The proof is simply to choose which unmarked circles carry x ; each such choice lowers degree by two.

Both counts can be computed with binary arithmetic without enumerating basis labels. The total count has ℓ bits. A direct factorial or multiplicative binomial routine has polynomial bit complexity in ℓ . The supplied closure helper checks matching involutions and noncrossingness, counts overlay circles, and returns these dimensions. It does *not* implement the action of general cobordism morphisms. The zero-boundary case is already a closed scalar stage and must be handled by its explicit reduced normalization rather than by inserting $\ell = 0$ in (5).

4.3 Rank is not structural reachability

A path in the transfer graph may be nonzero before closure and zero after closure. Several nonzero path products may cancel. For example, two disjoint unit-labelled paths from one source to one target give zero transfer over $\mathbb{F}_2$ even though their vertex cut has capacity at least one. In the dot algebra a path containing two copies of the same dot has zero product. The cut is a safe upper bound in all of these cases precisely because it makes no generic-coefficient assumption.

For a graded complex, construct a rank cap $\kappa_{h,j}$ for each $d^{h,j} : R^{h,j} \rightarrow R^{h+1,j}$. One may use a separate minimum cut in each sector, or reuse a separator with sector-specific weights. Reusing a

cut is sound but need not be optimal. The homology-budget calculation only requires valid upper bounds, not minimum cuts, so inexpensive heuristic separators can replace exact optimization without compromising correctness.

5 A sharp homology budget from adjacent rank constraints

5.1 Why summing all cut capacities is not enough

Let

$$c_{h,j} = \dim R^{h,j}, \quad r_{h,j} = \text{rank } d^{h,j}, \quad S = \sum_{h,j} c_{h,j}.$$

For a finite graded complex, exactness of the dimension formula gives

$$\beta := \sum_{h,j} \dim H^{h,j}(R) = S - 2 \sum_{h,j} r_{h,j}. \quad (7)$$

Rank caps yield $\beta \geq S - 2 \sum \kappa_{h,j}$, with zero as an additional trivial lower bound. A better local bound sums $\max(0, c_{h,j} - \kappa_{h-1,j} - \kappa_{h,j})$. Neither generally exploits all compatibility constraints. Since $d^2 = 0$,

$$r_{h-1,j} + r_{h,j} \leq c_{h,j}. \quad (8)$$

The same vector space cannot simultaneously supply unlimited incoming image and outgoing rank. Maximizing every individual rank cap independently may therefore overestimate how much homology can be cancelled.

Definition 5.1 (Optimized rank budget). Let $M(c, \kappa)$ be the maximum of $\sum_{h,j} x_{h,j}$ over nonnegative integers satisfying

$$0 \leq x_{h,j} \leq \kappa_{h,j}, \quad x_{h-1,j} + x_{h,j} \leq c_{h,j}. \quad (9)$$

All arrays have finite support and missing entries of c are zero. Define

$$\beta_{\text{cut}} = S - 2M(c, \kappa).$$

This is an unweighted capacitated matching problem on one path for each quantum degree. It is not a weighted matching objective and it is not a search over possible knot diagrams.

Theorem 5.2 (Optimal greedy budget and sharpness). *For each fixed j , scan the nonzero dimensions in increasing h , setting*

$$x_{h,j}^* = \min\{\kappa_{h,j}, c_{h,j} - x_{h-1,j}^*, c_{h+1,j}\}. \quad (10)$$

Treat a missing predecessor or successor as zero. This attains $M(c, \kappa)$. For every finite complex with these dimensions and rank caps,

$$\beta \geq \beta_{\text{cut}}. \quad (11)$$

Moreover, there exists a complex of vector spaces with exactly these chain dimensions, respecting the caps, whose total homology dimension is β_{cut} . Consequently no universally stronger bound can follow from these dimensions and rank caps alone.

Proof. The optimization separates by j and by gaps in the support of c . Consider the leftmost edge of a remaining degree path. Its maximum feasible value is $m = \min(\kappa_0, c_0, c_1)$. Take an optimal

feasible solution with first edge $x_0 \leq m$. Increase x_0 by $\Delta = m - x_0$ and, if necessary, decrease x_1 just enough to restore the constraint at the shared vertex. That decrease is

$$\max(0, m + x_1 - c_1) \leq \Delta,$$

because the old solution satisfied $x_0 + x_1 \leq c_1$. All other constraints remain satisfied: the left endpoint is respected by the definition of m , and decreasing x_1 cannot hurt its right endpoint. The objective does not decrease. Thus there is an optimal solution with first edge equal to m . Fix that value, subtract it from the capacity at the next vertex, and repeat. This proves (10) by induction.

For any actual differential, its ranks form a feasible solution by (8) and the given caps. Equation (7) then implies (11).

For sharpness, use the optimal integer allocation x^* . Decompose each vector space into a direct sum

$$R^{h,j} = B^{h,j} \oplus Z^{h,j} \oplus L^{h,j}$$

with dimensions respectively

$$x_{h-1,j}^*, \quad c_{h,j} - x_{h-1,j}^* - x_{h,j}^*, \quad x_{h,j}^*.$$

The middle dimension is nonnegative by feasibility. Map $L^{h,j}$ isomorphically to $B^{h+1,j}$ and map $B^{h,j}$ and $Z^{h,j}$ to zero. These maps square to zero, have ranks $x_{h,j}^* \leq \kappa_{h,j}$, and have total homology dimension $S - 2 \sum x_{h,j}^* = \beta_{\text{cut}}$. This realizes the claimed extremum over complexes of vector spaces. $\square$

The realization in the proof need not be a Khovanov complex and need not share the original transfer graph. This is precisely the scope of “sharp”: optimal information extraction from the numerical constraints, not realization of every feasible rank profile by a classical knot.

5.2 A strict improvement and binary-encoded dimensions

For a single quantum degree take

$$(c_0, c_1, c_2) = (2, 1, 2), \quad (\kappa_0, \kappa_1) = (1, 1).$$

The ungraded capacity sum gives lower bound one, and the sum of local lower bounds gives two. But $x_0 + x_1 \leq 1$, so $M = 1$ and

$$\beta_{\text{cut}} = 5 - 2 = 3.$$

The optimal calculation recognizes that the middle one-dimensional space cannot support both maximal ranks at once.

With sorted supports, the greedy calculation uses a constant number of comparisons, additions, and subtractions per nonzero dimension. Sorting costs $O(H \log(H + 2))$ index comparisons for H occupied bidegrees. If all counts and labels have at most B bits, a conservative bit bound is $O(H \log(H + 2) \text{ poly}(B))$. Large gaps in homological degree are not expanded. The same calculation applies when $c_{h,j}$ is given by a validated sum of binary-encoded multiplicities or closure binomial counts. Producing those counts and their validation evidence still has to be charged separately.

6 The rank-one decision dichotomy

Theorem 6.1 (Certified rejection or bounded terminal dimension). *Let R be a verified closed complex chain homotopy equivalent to the reduced Khovanov complex of a one-component classical knot. Suppose valid caps $\kappa_{h,j}$ have been constructed, and put $M = M(c, \kappa)$, $K_{\text{cap}} = \sum_{h,j} \kappa_{h,j}$. Then:*

1. *If $\beta_{\text{cut}} > 1$, the knot is nontrivial. No remaining transferred map needs to be evaluated to justify this conclusion.*
2. *If $\beta_{\text{cut}} \leq 1$, then*

$$S \leq 2M + 1 \leq 2K_{\text{cap}} + 1. \quad (12)$$

Thus the unresolved closed critical complex has bounded total dimension.

3. *In the second case, exact computation of the differential ranks and (7) decides the unknot: total rank one accepts, and a larger total rank rejects. A computed rank zero is an invalid knot-complex state, not an acceptance certificate.*

Proof. The first part follows from Theorem 5.2 and the reduced homology of the unknot. For the second, rearrange $S - 2M \leq 1$; the upper bound on M follows from the edge caps. The last part is the reduced binary rank-one criterion explained in Section 2. Every use of homology is justified by the assumed chain equivalence to the input knot complex. $\square$

This is a terminal dimension bound, not a kernelization theorem producing a small equivalent knot diagram. The graph before closure, the cost of its construction, the size of the scalar contraction, and the cost of verifying the closure functor may all remain large. We have removed a terminal multiplicity hypothesis in a *decision-specific* way, not proved a small state theorem at every intermediate crossing.

6.1 Why an inconclusive preflight must continue

A lower bound of one is not an upper bound of one. A concrete regression example has critical dimensions $(2, 1)$ and two identity pairs $L_1 \rightarrow B_1$, $L_2 \rightarrow B_2$. Each critical source has nonmatching arrows to both B_1, B_2 , and each L_i has a nonmatching arrow to the one critical target. After reversing the matched identities there are two paths from each source to the target. Their binary products cancel. The transferred differential is zero, so total homology dimension is three, while the target cut has capacity one and the preflight lower bound is one.

With just one identity pair the corresponding transferred row is $(1 \ 1)$, of rank one, and the total homology dimension is one. The same critical dimension summary and the same cut-capacity bound therefore permit different decisions. This example also rules out substituting a maximum flow value for the actual differential rank.

The implementation's preflight routine returns an algebraic `RANK_NOT_ONE` or an exact rank. It does not label an arbitrary complex `KNOTTED`. The braid-certificate wrapper establishes the knot-complex origin before returning topological verdicts.

6.2 Exact ranks without expanding the transfer matrix

After closure and, when needed, passage to one quantum sector, write

$$A : \mathbb{F}_2^a \rightarrow \mathbb{F}_2^\kappa, \quad B : \mathbb{F}_2^\kappa \rightarrow \mathbb{F}_2^b.$$

The dimensions here count vector-space basis elements, not formal matching objects. They need not equal the terminal and cut cardinalities in Proposition 3.3.

Proposition 6.2 (Rank through the interface). *A basis $u_1, \dots, u_t$ of $\text{im } A$, where $t \leq \kappa$, satisfies*

$$\text{rank}(BA) = \dim \text{span}\{Bu_1, \dots, Bu_t\}.$$

Classical binary elimination computes this rank in $O((a+b)\kappa^2)$ field operations, using $O(\kappa(a+b))$ bits for explicit factor storage and associated elimination work, up to the usual small-case conventions. It does not construct a $b \times a$ product matrix.

Proof. The image of BA is $B(\text{im } A)$, and a linear image of a spanning basis spans that image. Eliminating the a columns of a κ -row matrix costs $O(a\kappa^2)$ elementary field operations. Applying B to at most κ basis vectors costs $O(b\kappa^2)$, and eliminating their b -coordinate images has the same bound. Zero-dimensional factors have rank zero. $\square$

The reference code packs binary columns into Python integers. For typed rectangular-matrix coefficients it first flattens the factor blocks, keeping their row and column dimensions. This uses exact arithmetic and an explicit factor representation; there is no randomized projection or identity test. It deliberately exposes no “rank” over the dot algebra. Rank is taken only after applying a vector-space realization with verified types.

If $S \leq 2K_{\text{cap}} + 1$, summing the classical rank costs over all adjacent degrees gives $O((S+1)(K_{\text{cap}} + 1)^2)$ and hence a cubic envelope in $K_{\text{cap}} + 1$. The factors must first be constructed and evaluated; the next section keeps those costs separate.

7 Complete cost accounting and the quasi-polynomial goal

7.1 Parameters that must not be conflated

Let n be the crossing count of the original input diagram. For each transferred map use the following parameters:

Symbol	Meaning
V_i, E_i	Vertices and edges in its explicit typed transfer graph
r_i	Number of vertices in the chosen separator
κ_i	Sum of their closed vector-space dimensions
k_i	Half the tangle boundary size before closure
C_i	Cost of one exact typed coefficient operation
B	Bound on index, grading, dimension, and capacity bit lengths
S	Total closed critical dimension
T_{build}	Constructing the state, contraction, graphs, and their evidence
T_{close}	Computing closure dimensions and evaluating needed factors

Actual allocated slots count toward T_{build} unless a verified compact representation and its construction have been supplied. A compressed multiplicity is not a free oracle for its representative maps. Nor does small boundary width imply a small explicit complex.

7.2 A conservative implementation-level bound

Theorem 7.1 (Terminal complexity with all interfaces charged). *Assume explicit valid transfer graphs and an evaluable fixed closure as above. With the reference Edmonds–Karp cut search, a conservative bit-cost envelope for the terminal rank-one algorithm is*

$$T \leq T_{\text{build}} + T_{\text{close}} + \text{poly}(B) \sum_i O(V_i(E_i + V_i + 1)^2) + \text{poly}(B)O(H \log(H + 2)) \\ + \mathbf{1}_{\{\beta_{\text{cut}} \leq 1\}} \left[\sum_i O(r_i(V_i + E_i + 1)) (C_i + \text{poly}(B)) + \text{poly}(B)O((S + 1)(K_{\text{cap}} + 1)^2) \right], \quad (13)$$

with topological-sorting index work added as in Proposition 3.3. Closure evaluation costs in the exact branch are included in T_{close} ; rejection need only pay the closure work necessary to certify dimensions and rank caps. A supplied, independently verified cut can replace minimum-cut search.

Proof. Node splitting produces $O(V_i)$ vertices and $O(E_i + V_i)$ arcs. Edmonds–Karp uses $O(V_i(E_i + V_i + 1)^2)$ arithmetic operations [2]. Residual capacities have polynomially bounded bit length in the input capacities and graph size. Direct verification of a supplied cut and matching flow is linear in this network size apart from binary arithmetic. The greedy budget has the cost established in Section 5.

When the lower bound rejects, these operations suffice. Otherwise construct the first-hit factors using Proposition 3.3, evaluate the needed closure blocks, and use Proposition 6.2. If the closed source and target dimensions of map i are a_i, b_i , then

$$\sum_i (a_i + b_i) \kappa_i^2 \leq 2S \sum_i \kappa_i^2 \leq 2SK_{\text{cap}}^2.$$

Each chain group occurs in at most two adjacent maps, and all quantities are nonnegative. This proves the rank term and the displayed envelope. $\square$

The expression is deliberately not a wall-clock forecast. Python hashing, object allocation, and dense bit columns affect constants; deterministic array or ordered-map implementations add controlled index overhead. A production adapter also needs cancellation, time, and memory polling. The reference implementation is small and exact, not an implementation of such a production resource policy.

Corollary 7.2 (Conditional quasi-polynomial recognition class). *Consider a class of n -crossing diagrams for which an exact front end constructs and verifies the required terminal transfer input in $\exp(O(\log^2(n + 2)))$ bit time. Suppose closure work and the number of maps have that same bound, all explicit graph sizes and separator capacities are at most $\exp(O(\log^2(n + 2)))$, coefficient operations have cost at most $\exp(O(\log^2(n + 2)))$, and label bit lengths are at most $\exp(O(\log^2(n + 2)))$. Then the complete construction followed by the terminal algorithm has quasi-polynomial bit complexity on that class. A sufficient coefficient hypothesis in the maintained cobordism setting is $k_i = O(\log^2(n + 2))$ together with a justified $C_i = 2^{O(k_i)} \text{poly}(k_i)$ representation bound.*

Proof. Every term in (13) is a fixed-degree polynomial in the stated quasi-polynomial quantities. On the unresolved branch, Theorem 6.1 bounds S by $2K_{\text{cap}} + 1$. A fixed-degree polynomial in $\exp(O(\log^2(n + 2)))$ has the same asymptotic form. The stated coefficient envelope has that form under the boundary hypothesis. Recognition follows from the knot-complex contract. $\square$

The construction hypothesis is substantive and unproved for unrestricted input diagrams. The supplied cube front end is exponential, and the conditional corollary does not make it quasi-polynomial. Furthermore, if the full critical basis has already been expanded, observing a small rank budget after the fact does not refund its construction cost. A useful next theorem must control the *production* of the graph and dimension summary, not merely their consumption.

7.3 What changes relative to an endpoint-only analysis

Endpoint propagation can bound a degree map by roughly $|E_i| \min(a_i^{\text{obj}}, b_i^{\text{obj}})$ coefficient operations, with refined reachability counts when available. First-hit propagation uses internal separator cardinality instead of either terminal cardinality, and terminal rank can be computed through the smaller closed interface. The cut-only budget test adds another branch: sometimes no coefficient propagation is needed at all. These are genuinely different cost expressions, but none uniformly dominates an optimized sparse cancellation strategy.

The dimension dichotomy also distinguishes total homology computation from unknot recognition. A family can have very large homology and still be easy to reject from a small rank cap. Consequently lower bounds or output-size obstructions for computing the entire invariant cannot simply be transferred to this rank-one decision query.

8 A strict algebraic separation from both propagation directions

8.1 An explicit family of valid complexes

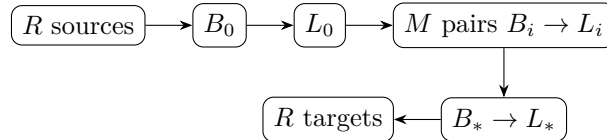
Fix positive integers R, M . Let the critical degree-zero generators be $s_1, \dots, s_R$ and the critical degree-one generators be $t_1, \dots, t_R$. Add degree-zero generators $L_0, L_1, \dots, L_M, L_*$ and degree-one generators $B_0, B_1, \dots, B_M, B_*$. Set

$$d_0(L_i) = B_i \quad (0 \leq i \leq M), \quad d_0(L_*) = B_*.$$

Add the following unit-coefficient perturbation entries:

$$s_j \longrightarrow B_0, \quad L_0 \longrightarrow B_i \quad (1 \leq i \leq M), \quad L_i \longrightarrow B_*, \quad L_* \longrightarrow t_j.$$

There are no other maps. This is a two-term complex, so $d^2 = 0$ immediately. The displayed identity pairs form an acyclic Morse matching. Reversing them gives the transfer graph sketched below; all quantum labels can be zero in this closed binary model.



Every source–target entry is the sum of M unit path products. Thus for odd M the transferred matrix is the all-ones $R \times R$ matrix, with rank one. The closed critical homology dimension is $2R - 2$.

Proposition 8.1 (Internal-cut separation). *For odd M , this graph has*

$$|V| = 2R + 2M + 4, \quad |E| = 2R + 3M + 2.$$

The endpoint reference evaluates $R(3M+R+3)$ coefficient edges in either forward or reverse direction. First-hit factorization at $\{B_0\}$ uses $3M + 2R + 2 = |E|$ coefficient compositions and produces a rank-one factorization. Its minimum cut has capacity one. Taking even R and $M = R^2 + 1$ gives $\Theta(R^3)$ endpoint propagation work versus $O(R^2 \log(R + 2))$ total index-word work for cut discovery, factor construction, and rank, with $O(R^2)$ stored graph words. The terminal budget rejects without evaluating the factors for every $R \geq 2$.

Proof. The vertex and edge counts follow by listing the $2R$ critical vertices, $2(M + 2)$ matched vertices, $2R + 2M$ nonmatching edges, and $M + 2$ reversed identity edges. For a fixed source, forward propagation traverses its first edge, the next identity, the M outward nonmatching arrows, the M reversed identities, the M inward nonmatching arrows, the final identity, and the R output arrows. Since M is odd, the value at B_* is nonzero. The count is $3M + R + 3$. Symmetry gives the same reverse count per target.

The first-hit prefix at B_0 traverses R edges. Its unrestricted suffix traverses $1 + M + M + M + 1 + R$ edges. The total is $3M + 2R + 2$. The one-dimensional vertex B_0 separates all paths. At least one path exists, so a unit integral flow certifies minimum capacity one. Edmonds–Karp performs one augmentation and a final unsuccessful search on this family. Topological sorting and indexed representation therefore cost $O((R + M) \log(R + M + 2))$ word operations; factor evaluation is linear in the graph size. Rank of the one-column/one-row factorization costs $O(R)$. For $M = R^2 + 1$ these bounds give the claimed separation. Finally $S = 2R$ and $M(c, \kappa) = 1$, so $\beta_{\text{cut}} = 2R - 2 > 1$ for $R \geq 2$. $\square$

This is an exact asymptotic separation between specific complete rank-stage implementations on valid algebraic complexes. It is not a new lower bound for every algorithm computing their rank. Sparse Gaussian cancellation can also exploit the shared structure efficiently. No realization of this family by classical knot prefixes is proved, and its even total homology dimension makes a direct reduced-knot interpretation particularly inappropriate. The role of the family is to expose and test the internal sharing that both endpoint choices repeat.

9 Implementation, certificates, and executed validation

9.1 Module boundaries

The package uses only Python’s standard library. `core.py` implements typed DAG validation, three coefficient adapters, endpoint propagation, first-hit factors, factorized binary rank, weighted minimum cuts, and an independent cut/flow verifier. `budgets.py` implements Theorem 5.2. `closure.py` checks marked matching dimensions, not cobordism maps. `decision.py` implements the actual cut-only preflight and exact fallback.

`cube.py` constructs a small reduced binary Khovanov cube for a closed braid, validates grading and $d^2 = 0$, constructs a deterministic partial acyclic matching, and exports the resulting scalar transfer graphs. It is intentionally exponential and rejects input beyond eleven crossings. Its matching-edge limit bounds the number of candidate edges considered; it is not the number of matched pairs. No claim about optimal matchings is made. The experiment uses limits 12 and unbounded on each small diagram.

The topological coefficient conventions are written out in Appendix A. They provide a checkable bridge to real knots, rather than merely declaring random graded matrices to be knot data. Still,

this small cube constructor is not an independent established knot-software oracle. The test suite also checks known examples and elementary braid transformations to exercise that bridge.

9.2 Portable braid certificates and verifier independence

`certificates.py` produces certificates containing the input braid, a deterministic hash of its reconstructed cube, the matching pairs, per-degree cut vertices and feasible flows, transferred ranks, and the final reduced rank and verdict. The verifier reconstructs the cube, checks the matching and graph acyclicity, verifies each cut/flow pair, and recomputes transferred ranks by *dense endpoint propagation*. It does not call the matching search, minimum-cut optimizer, first-hit factorizer, or factorized-rank routine.

This is useful implementation independence, not independence from all mathematical assumptions. Producer and verifier share the cube constructor, coefficient conventions, and generic binary-rank primitive. The verifier is not polynomial in the original crossing count, and the certificate format is not offered as a new complexity-class certificate. Hashes bind files and reconstructed inputs; they do not prove topology.

Six braid certificates are supplied and were replayed successfully. Mutation tests alter rank, input word, matching, or cut capacity and require rejection. A separate mocked test makes factor evaluation raise an exception and confirms that a cut-only rejection does not call it. Another test has lower bound one and exact rank three, guarding against unsound early acceptance.

9.3 Unit tests and cross-checks

The executed unit-test command is

```
PYTHONPATH=code python -m unittest discover -s tests -v
```

It passed 36 tests. These test methods contain multiple exhaustive or seeded subcases: valid separators in random small DAGs, all weighted cuts in small graphs, 500 brute-force budget comparisons, and 40 additional random knot closures, among others. They are not counted as separate unit tests.

The categorical comparisons cover $\mathbb{F}_2$, the square-zero dot algebra, and typed rectangular matrices. They include non-antichain cuts, cut terminals, empty separators when no terminal path exists, cancelled path sums, nilpotent products, and noncommuting compositions. Weighted-cut tests include zero and thousand-bit capacities. Graded budget tests include missing degree intervals and dimensions whose labels must not be expanded. The cube tests include mirrors, cancelling inverse generators, the three-strand braid relation, and positive and negative Markov stabilization on small examples. These finite tests support the implementation; the general correctness claims rest on the proofs, not on statistical extrapolation.

9.4 An 80-diagram knot-complex audit

The seeded corpus consists of eight named or elementary braid examples and 72 random one-component braid closures. Random diagrams use two to four strands and two to seven crossings; words are retained verbatim in the data. The same knot may occur repeatedly under different diagrams. No claim of 80 distinct knot types or of an adversarial hard-knot corpus is intended.

For every diagram we construct the full reference cube and its independently eliminated bigraded homology, then compare two Morse configurations. The result is 160 configurations and 1,070 transferred maps, with no entrywise transfer, factor-rank, or bigraded homology disagreement. Across the 80 original cubes there are 24,556 allocated generators.

Twenty-four diagrams have reduced rank greater than one. All 24 are rejected by cut-only preflight in both matching configurations, giving 48 rejections without transferred-map evaluation. This does not mean that their prior cube construction was avoided. The sharp bound strictly exceeds the naive global capacity bound in 52 configurations and strictly exceeds the sum of local nonnegative bounds in 51. These are finite-corpus observations, not a general detection-completeness theorem for preflight.

Named diagram	Crossings	Cube generators	Reduced rank	Full-match critical
Crossingless unknot	0	1	1	1
Positive RI unknot	1	3	1	1
Negative RI unknot	1	3	1	1
Trefoil	3	15	3	3
Figure-eight	4	33	5	5
$T(2, 5)$	5	123	5	5
Stabilized unknot	2	9	1	1
Cancelled-generator unknot	3	15	1	1

10 Measurements: speedups, overhead, and what was not timed

10.1 Paired protocol

All arms answer the same *exact differential-rank query* on the same preconstructed graph or collection of degree graphs. The forward and reverse arms construct the full endpoint transfer matrix and take its binary rank. The cut-factor arm includes network construction, exact minimum-cut search, cut/flow verification, first-hit factors, and binary rank through those factors. The control arm is identical to forward.

Every arm is warmed once. Seven rounds then execute a seeded shuffled arm order, with garbage collection before each timing and fresh algorithm-local arrays in each call. Timings use `perf_counter_ns`; medians are reported. These short local timings are not hardware-independent constants. The raw round order, individual values, Python version, platform, corpus, and counts are retained in `results/`.

Graph construction, cube construction, and Morse matching are excluded from *every* timed arm. No production `fastunknot` scanner, minimum-fill sparse reducer, recognition filter, ordering heuristic, or Rust port is benchmarked. The cut-only decision shortcut is validated separately, not substituted for exact rank in the timing comparison.

10.2 Small actual knot complexes: discovery overhead dominates

The following medians are milliseconds for all transfer maps of the named partial-matching configuration. Values below one millisecond should be interpreted with the accompanying A/A control, not as precise universal ratios.

Case	Forward	A/A control	Reverse	Cut-factor	F/C
trefoil	0.054	0.056	0.055	0.130	0.41
figure_eight	0.125	0.127	0.129	0.455	0.27
torus_2_5	0.497	0.480	0.478	2.130	0.23

The cut-factor method is slower by approximately $2.41\times$, $3.64\times$, and $4.29\times$ respectively against forward propagation. The graphs and endpoint sets are too small for internal sharing to repay network and verification overhead. This is evidence against changing the default reducer on the basis of this package, not a reason to conceal the measurements.

10.3 Constructed bottleneck family: an increasing separation

The next table records exact composition counts for Proposition 8.1. In every row the forward and reverse counts coincide. The first-hit count equals the graph’s edge count.

R	M	Vertices	Edges	Endpoint compositions	First-hit compositions
8	65	150	213	1648	213
16	257	550	805	12640	805
32	1025	2118	3141	99520	3141
64	4097	8326	12421	790912	12421

At $R = 64$, first-hit propagation uses 12,421 compositions instead of 790,912, a count reduction of approximately $63.68\times$. This is not the wall-clock speedup because both graph work and minimum-cut discovery are included in the complete cut-factor arm.

Case	Forward	A/A control	Reverse	Cut-factor	F/C
$R = 8$	0.528	0.536	0.540	0.790	0.67
$R = 16$	3.394	3.366	3.417	2.750	1.23
$R = 32$	25.819	24.758	26.733	10.897	2.37
$R = 64$	191.937	192.894	189.887	39.957	4.80

At the smallest size discovery overhead still wins against factorization. At the largest size the complete cut-factor rank stage is about $4.80\times$ faster than forward propagation and $4.75\times$ faster than reverse propagation. The A/A medians are also shown. These results support the proved sharing mechanism on an algebraic stress family. They neither establish a speedup on difficult classical knots nor compare against the maintained sparse-cancellation default, which could exploit the same family in a different way.

10.4 Reproduction and interpretation

Run the full driver with

```
PYTHONPATH=code python code/run_experiments.py
```

It reconstructs the seeded corpus, verifies exact equalities, runs the paired experiments, writes raw JSON and CSV, and regenerates the six example certificates. Re-running it changes timing samples,

not the expected exact counts or homology comparisons. The numerical tables in this PDF are a frozen rendering of the delivered run; the supplied table generator updates them from a new result file. Time comparisons should be based on paired results from one environment rather than copied between machines.

11 Integration plan and safety of topological verdicts

11.1 An additive, opt-in integration boundary

A suitable first integration is a research subdirectory under `Topology/UnknotRecognition/reports/closure_cut_transfer/`, preserving the maintained recognizer unchanged. No numeric report identifier is assumed available. The independent code can be imported or run from that directory, and the article can be linked from the synthesis overview after review.

A production implementation should first export a validated degree graph from the existing graded/corridor transfer boundary and compare every expanded first-hit map against the existing implementation. It should then attach a verified fixed-closure adapter and compare graded closed matrices. Only after these two bridges are established should the cut-only homology budget be permitted to produce a recognition verdict.

It is especially important not to feed the new rank-only answer back into an unfinished tangle scan. A rank does not encode the transferred morphism, and the future continuation can distinguish maps of equal rank before closure. Full typed factors can remain useful for further algebra only if all subsequent operations are defined on that representation and preserve its semantics. That is a separate implementation and proof task.

11.2 Transactional and resource obligations

The reference code rejects cyclic graphs, invalid cuts, malformed basic encodings, mismatched types encountered in composition, and link inputs to the knot-certificate wrapper. This is not a complete hostile-input or resource-management layer. A real adapter must validate all coefficients and endpoint types before trusting graph-derived data, account for actual allocated arrays, poll cancellation/deadline limits during flow and algebra, and preserve the previous valid state if installation fails.

A safe production policy builds a prospective replacement and its evidence without mutating the live complex, verifies all dimensions and maps needed for the selected branch, checks the remaining budget, and only then commits an exact state or a certified verdict. Failure to finish becomes `UNKNOWN` or an exact fallback on the preserved input. It is never interpreted as knottedness, triviality, or a rank bound.

Cut search itself can be optional. A separator supplied by a crossing order, a previously computed corridor, or an inexpensive heuristic can be verified and used directly. Exact minimum-cut search is relevant when its stronger cap repays its overhead. No competitive scheduling theorem for choosing among sparse cancellation, endpoint propagation, and internal cuts is proved here.

11.3 Proof and engineering checklist before promotion

The contract must bind a graph to a particular input knot, matching/scalar contraction, boundary marking, grading convention, and fixed closure. Certificate checks must establish acyclicity, scalar contraction validity, closure dimensions, and coefficient types. For terminal acceptance, the complete rank calculation must be covered; a narrow quantum or homological window that merely agrees with the unknot is not sufficient. For rejection, one certified lower bound above one is sufficient, but every assumption supporting that lower bound must still be checked.

Finally, benchmarks intended to justify a default change must include the full recognizer, existing filters, order selection, diagram preparation, resource handling, and all proposed adapter costs. The present data isolate one research boundary and intentionally stop short of that claim.

12 Further research questions and proposed next results

12.1 Closure-aware separator profiles on genuine hard inputs

For an actual verified transfer graph, measure the profile

$$(h, j) \mapsto (V_{h,j}, E_{h,j}, a_{h,j}, b_{h,j}, r_{h,j}^{\text{cut}}, \kappa_{h,j}).$$

Here r^{cut} denotes cut cardinality, not differential rank. The first experimental question is whether difficult classical knot diagrams ever have small *weighted* internal cuts that are not already handled well by sparse elimination or endpoint-direction choice. A useful study should include diagrams surviving the existing recognition filters, not merely a random small braid corpus.

The proposed deliverable is an exact exporter plus frozen actual-stage fixtures, with independent contraction and closure checks. A theorem about one explicit infinite classical-knot family would be stronger evidence than additional unrelated algebraic stress examples.

12.2 Finding useful cuts without paying for exact flow

Can a separator derived from the quantum order, a shared prefix, a shared suffix, or a small number of articulation regions achieve capacity within a provable factor of the minimum? A verified nonminimum cut is already sound. The algorithmic objective should combine rank capacity, cut cardinality, closure-evaluation cost, and expected propagation work; minimizing just one of those quantities may choose an expensive representation.

A precise target is an approximation theorem for a weighted objective of the form $\alpha\kappa + \gamma r + \sum_{q \in Q} \text{evalcost}(q)$ on the restricted DAGs arising from valid contractions. Any such theorem needs its own proof: the ordinary unweighted path-budget greedy rule does not solve this separator optimization.

12.3 Basis-aware minimization of transfer interfaces

Scalar basis changes alter I, P, H and hence the corridor graph without changing the represented homotopy type. Is there an efficiently certifiable choice of scalar contraction that decreases a closure-capacity profile or its sum? An unconstrained optimization over bases is too broad. One promising restricted problem permits basis changes only inside fixed matching and quantum blocks, with bounded-support representatives.

The desired result would state both the basis-search cost and the resulting factor sparsity. A small cut obtained only after a dense exponential basis change would not improve recognition complexity.

12.4 Stronger numerical certificates using additional relations

The budget theorem is sharp only given dimensions and separate rank caps. Known zero maps, grading support, vanishing composites, symmetry, or shared factor spaces can carry additional information. Can one incorporate such information while retaining a polynomial-time certificate optimizer? A carefully chosen extension might improve rejection without computing ranks, but arbitrary matrix-rank constraints should not be assumed to reduce to path matching.

A specific candidate is to record a common interface for several adjacent maps together with a verified relation between their images and kernels. The challenge is to formulate constraints stronger than $x_{h-1} + x_h \leq c_h$ without guessing algebraic independence.

12.5 Succinct multiplicities and the cost of knowing dimensions

The dichotomy is most useful when S can be computed before allocating all S basis vectors. Which maintained shared-component or twist representations admit exact binary-encoded $c_{h,j}$ and a verifiable closure action on the necessary maps? A useful theorem would prove polynomial-time extraction of these counts in the compressed representation size and would bound the factor dimensions required only in the unresolved branch.

The principal obstruction is that repeated objects can share dimensions without sharing all incident morphisms. A representation must encode the maps and their interactions, not simply replace equal matching labels by a multiplicity counter.

12.6 Persistence under crossing extension

Can first-hit factors be updated after one crossing is appended without expanding the old product? To obtain a general scanner improvement, one needs formulas for extension, direct sums, delooping, and cancellation on the factored representation, together with size bounds over many stages. A small factorization at one terminal graph does not itself persist under these operations.

A realistic intermediate theorem would treat a fixed number of prescribed continuation types and establish bounded factor growth under that restricted alphabet. A universal logarithmic-depth or quasi-polynomial-size statement would require genuinely new geometric or categorical control.

12.7 Decision-first certificates beyond rank one

For any threshold b , the same proof gives the algebraic dichotomy

$$\beta_{\text{cut}} > b \quad \text{or} \quad S \leq 2M + b.$$

Can thresholded computations accelerate other link invariants or certified homology queries after the topological meaning of the threshold is fixed? This extension is algebraically immediate; its usefulness and appropriate normalization for links are not. The one-component unknot contract must not be silently reused for arbitrary link thresholds.

12.8 Formal verification and certificate minimization

The typed first-hit theorem, cut verification, and greedy rank-budget proof are small enough to be plausible early formalization targets. A next artifact could formalize finite-DAG path decomposition and the exchange proof in Lean, then connect the statements to the existing matching or cobordism structures. The present package contains no Lean files and claims no formal verification.

The example braid certificates deliberately use a dense independent verifier. A useful engineering improvement would reduce verifier work while retaining independence, for example by supplying checked basis and image certificates for $\text{im } A$ and $B(\text{im } A)$. That would improve certificate replay, not eliminate the need to bind the factors to a genuine knot complex.

12.9 A class theorem with an affordable construction

The most consequential missing result is an explicit class of difficult classical diagrams for which the *construction*, closure work, graph sizes, and capacity budget in Corollary 7.2 are all bounded in terms of the input size. Logarithmic boundary width alone is insufficient. The theorem should identify a geometric condition that can itself be checked or supplied with bounded-size evidence.

A successful class theorem would turn the current terminal algebra into an end-to-end asymptotic improvement. Without that step, further refinement of matrix evaluation alone does not establish the target for unrestricted knots.

12.10 Relationship with hierarchy-based recognition

A homological terminal certificate and a geometric hierarchy certificate can be complementary exact branches in a recognizer. Any proposed portfolio should retain independent provenance and costs for both branches. The new cut theorem supplies no bound for constructing hierarchy surfaces, compressed cutting, or the cost of essential-pattern tests. Conversely, a geometric simplification that constructs a small verified terminal tangle might make our capacity analysis useful. Finding such a precise interface is a research question, not an equivalence asserted here.

13 Conclusion and claim ledger

The internal interface matters in two different ways. It can share a large path sum more efficiently than either endpoint direction, and its closed dimension can limit how much differential rank is available to remove critical homology. The latter effect leads to the sharp budget theorem and a rejection-or-small-terminal-dimension dichotomy. These are exact algebraic statements, and the implementation exercises them both on genuine small knot complexes and on a controlled algebraic separation family.

The main bottleneck that remains is not an algebraic identity left unchecked. It is the construction and maintenance of an affordable, topologically verified representation on arbitrary inputs. The numerical evidence also argues for an optional research integration rather than a default replacement: cut discovery loses on the small actual knot rank stages tested here.

Claim	Status
First-hit factorization for arbitrary typed DAG cuts	Proved; categorical tests passed
Closure-capacity rank upper bounds	Proved under a verified functor contract
Optimal budget and sharpness from dimensions/caps	Proved; exhaustive small comparisons passed
Rank-one rejection/dimension dichotomy	Proved for certified closed knot complexes
Independent small-braid certificate replay	Implemented; six supplied certificates verified
General production cobordism closure adapter	Not implemented in this package
Full maintained upstream test suite and recognizer benchmarks	Not run here
Quasi-polynomial algorithm for arbitrary input diagrams	Not established here

The contribution is therefore a proof-supported research extension with reproducible validation, explicit asymptotic scope, and concrete next theorem targets, rather than an unverified claim of a general complexity breakthrough.

A Reference reduced-cube conventions

A braid word on s strands is an integer list $w_1, \dots, w_n$ with $1 \leq |w_i| < s$. The permutation obtained by swapping positions $|w_i|, |w_i| + 1$ at each generator determines the number of closure components. The knot-certificate wrapper requires one cycle. For each crossing, positive zero-resolution means vertical arcs and positive one-resolution means cap-cup; the negative generator swaps those choices. Union-find computes resolution circles after identifying the top and bottom of each strand. The circle containing the first closure strand is marked.

The Frobenius algebra is $A = \mathbb{F}_2[x]/(x^2)$ with

$$m(1, 1) = 1, \quad m(1, x) = m(x, 1) = x, \quad m(x, x) = 0,$$

$$\Delta(1) = 1 \otimes x + x \otimes 1, \quad \Delta(x) = x \otimes x.$$

We use the marked subcomplex in which the marked circle always has label x , and shift its reduced grading by $+1$. With state σ , let $|\sigma|$ be its number of one-resolutions, $c(\sigma)$ its number of circles, and u the number of x labels including the mark. The normalized bidegree is

$$h = |\sigma| - n_-, \quad j = |\sigma| + n_+ - 2n_- + c(\sigma) - 2u + 1.$$

A cube edge raises h by one and preserves j . Binary coefficients avoid cube signs. The code checks these statements for every constructed edge and checks $d^2 = 0$ on every generator before using the complex.

The reference construction enumerates all 2^n resolutions and all reduced circle labellings. Both memory and work can be exponential. The eleven-crossing cap is an explicit safety limit on this research utility, not a theorem that larger diagrams are difficult or that exhausting the cap has any mathematical meaning.

B Algorithm summaries and certificate formats

B.1 First-hit transfer

```

check graph is a finite typed DAG
check Q separates all sources from all targets
for q in Q:
    A[q, :] = reverse_propagation_from(q,
                                       stop_at_other_vertices_of_Q=True)
    zero A[q, s] for every s in Q with s != q
    B[:, q] = unrestricted_forward_propagation_from(q)
return (A, B) # exact identity D = B*A

```

A stopped vertex may receive a value, but it must not propagate that value or be included as another cut source's first-hit contribution. Identities at cut terminals represent length-zero paths. These conventions account for the terminal-cut regression cases.

B.2 Decision preflight

```

verify input complex/contraction/closure contract
compute closed critical dimensions c[h,j]
for every adjacent-degree transfer graph:
    find or receive a vertex separator
    verify the separator and its closure-weighted capacity
compute M = maximum compatible sum of ranks by the greedy rule
lower = sum(c) - 2*M
if lower > 1:
    return certified rank-not-one # knot rejection only with origin contract
otherwise:
    compute exact transferred ranks through first-hit factors
    beta = sum(c) - 2*sum(exact_ranks)
    return exact beta

```

The prototype uses unit capacities on already evaluated scalar graphs. The theorem also covers weighted typed graphs once a verified closure adapter provides their dimensions and maps.

B.3 Cut certificate

A cut certificate stores the cut-vertex list, its capacity, and one flow integer per arc of a canonical node-split network. Verification checks: nonnegative capacities and flows; every flow within its capacity; conservation at all nonterminal network vertices; the asserted total value; and absence of a surviving source–target path after deleting the cut. Equality of flow value and cut capacity proves optimality. For a nonminimum but valid rank cap, only the separation and dimension checks are necessary.

B.4 Braid-certificate commands

From the package root:

```
PYTHONPATH=code python -m separator_transfer braid \
  --strands 3 --word '[1,-2,1,-2]' \
  --certificate examples/my_figure_eight.json
PYTHONPATH=code python -m separator_transfer verify \
  examples/my_figure_eight.json
```

These commands are reference computations on a small diagram. They do not call the maintained production recognizer. User-supplied files can be modified arbitrarily, so a verifier must rebuild and check the required mathematical data rather than trusting the verdict field.

C Artifact map and reproducibility scope

Path	Purpose
article/article.tex, references.bib	Complete article and primary references
article/article.pdf	Compiled and visually inspected report
code/separator_transfer/	Exact transfer, flow, budget, cube, and certificate modules
code/run_experiments.py	Seeded audit and paired benchmark driver
tests/test_research.py	36 unit tests with exhaustive and seeded subcases
results/knot_audit.json	Every diagram, matching configuration, and exact result
results/benchmarks.json	Raw paired samples and measurement scope
results/summary.json	Auditable summary counts and medians
examples/	Six independently replayed braid certificates
PROVENANCE.json, INTEGRATION.md	Reviewed-source identifiers and adapter obligations
SHA256SUMS	Hashes of delivered files other than the manifest itself

The package includes no font files, third-party paper PDFs, or copied production source. It is intended to be reviewed and integrated as an additive research artifact. Exact results are deterministic under the stated encoding and seed; timing values are observations from the delivered run.

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